

Effect of nicotinamide riboside on airway inflammation in COPD: a randomized, placebo-controlled trial

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Chronic obstructive pulmonary disease (COPD) is a progressive, incurable disease associated with smoking and advanced age, ranking as the third leading cause of death worldwide. DNA damage and loss of the central metabolite nicotinamide adenine dinucleotide (NAD⁺) may contribute to both aging and COPD, presenting a potential avenue for interventions. In this randomized, double-blind, placebo-controlled clinical trial, we treated patients with stable COPD ($n = 40$) with the NAD⁺ precursor nicotinamide riboside (NR) for 6 weeks and followed-up 12 weeks later. The primary outcome was change in sputum interleukin-8 (IL-8) from baseline to week 6. The estimated treatment difference between NR and placebo in IL-8 after 6 weeks was -52.6% (95% confidence interval (CI): -75.7% to -7.6% ; $P = 0.030$). This effect persisted until the follow-up 12 weeks after the end of treatment (-63.7% ; 95% CI -85.7% to -7.8% ; $P = 0.034$). For secondary outcomes, NR treatment increased NAD⁺ levels by more than twofold in whole blood, whereas IL-6 levels in plasma remained unchanged. In exploratory analyses, treatment with NR showed indications of upregulated gene pathways related to genomic integrity in the airways and reduced epigenetic aging, possibly through a reduction in cellular senescence. These exploratory analyses need to be confirmed in future trials. ClinicalTrials.gov identifier: [NCT04990869](https://clinicaltrials.gov/ct2/show/study/NCT04990869).

Aging is among the most important risk factors for developing chronic diseases¹, and chronic obstructive pulmonary disease (COPD) has been described as a disease of accelerated lung aging². Nicotinamide adenine dinucleotide (NAD⁺) has emerged as a central molecule involved in multiple age-related pathways, including immune function and inflammation³. Importantly, NAD⁺ modulates DNA repair capacity⁴ and decreases in some tissues with aging in humans, in various animal models⁵ and in premature aging diseases⁶. Nicotinamide

riboside (NR) is an NAD⁺ precursor that mitigates aging dysfunction⁷, and even short-term treatment with NR rescues DNA repair disorders in animal models⁶. In humans, modulating NAD⁺ levels may improve insulin resistance⁸ and some features of Parkinson's disease⁹. Although the results from some studies have been discouraging^{10–12}, the anti-inflammatory effect of NR is the most consistent finding thus far¹³. The effect of augmenting NAD⁺ levels in COPD has not yet been investigated.

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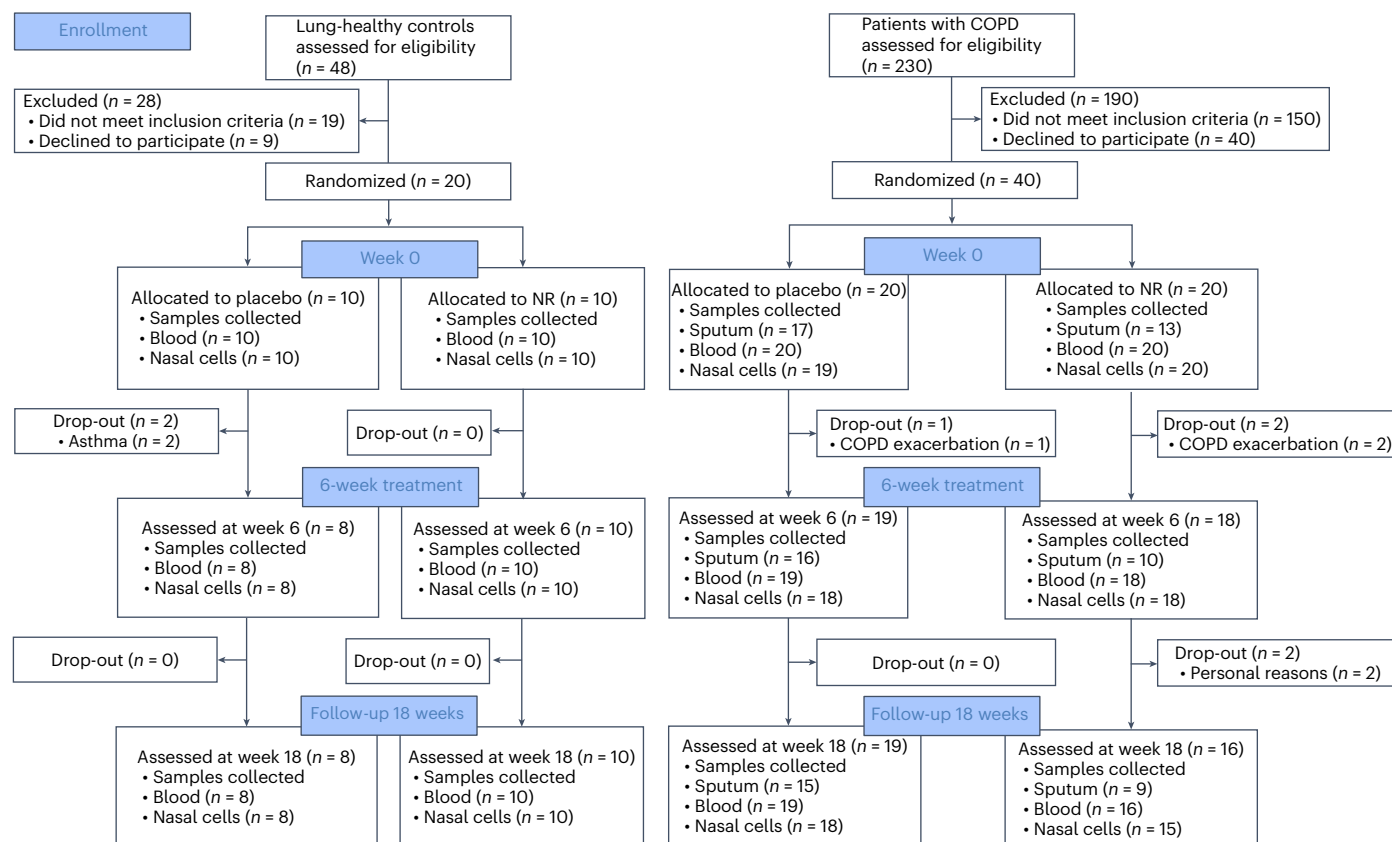


Fig. 1 | CONSORT flow diagram. The primary outcome IL-8 was assessed in sputum samples from patients with COPD.

In the present study, we hypothesized that boosting NAD⁺ levels in previously smoking patients with COPD would reduce airway inflammation and cellular senescence and lead to an upregulation of transcription pathways related to DNA repair. To investigate this, we conducted a randomized, double-blind, placebo-controlled clinical trial in which patients with non-eosinophilic COPD and lung-healthy controls were supplemented daily with NR (1 g, twice a day) or placebo for 6 weeks. Because interleukin-8 (IL-8) is part of the senescence-associated secretory phenotype (SASP)¹⁴ and a driving factor of neutrophilia in COPD¹⁵, this cytokine was chosen as the primary outcome of the trial.

Results

Patient characteristics and safety

Forty patients with COPD (mean age, 71.9 years) and a blood eosinophil count of less than 0.3×10^9 cells per liter, who were ex-smokers and had a smoking history of more than 10 pack-years, were recruited into the study. Furthermore, we enrolled a convenience sample of lung-healthy controls who were similar to the patients with COPD in terms of age, sex and body mass index (BMI) but who reported being never-smokers with no history of lung disease (mean age, 70.9 years). Participants were randomized 1:1 to receive either NR or placebo for 6 weeks and attended three study visits: pre-treatment and post-treatment and a follow-up 12 weeks after post-treatment. Patients with COPD were randomized and stratified by COPD Assessment Test (CAT) score (0–15 and 16–40)¹⁶. One patient with COPD in the placebo group and two in the NR group dropped out before attending the post-treatment visit, and an additional two patients in the NR group dropped out before the follow-up visit (Fig. 1).

Baseline characteristics of participants are included in Table 1. Patients with COPD and lung-healthy controls were pooled into their respective treatment groups to assess adverse events (AEs). A total of 29 AEs were reported by 26 participants, and there was no difference

when comparing NR to placebo either in patients with COPD or in lung-healthy controls (Extended Data Table 1). Previous studies indicated that NR may cause gastrointestinal issues^{17,18}; however, we found that only 10% of participants receiving NR and 14% receiving placebo experienced such issues. No serious AEs were observed. Thus, a relatively high daily dose of oral supplementation with NR was found to be safe and well tolerated in older individuals with and without COPD who were enrolled in our study.

Primary outcome IL-8 in sputum

We first evaluated the effect of NR supplementation on airway inflammation in sputum—representing central airway cell populations¹⁹—collected in patients with COPD (Fig. 2a). The primary outcome measure of the trial was the chemokine IL-8, which is an essential aggravator of cellular senescence¹⁴, recruits inflammatory neutrophils to the lung of patients with COPD¹⁵ and is related to disease severity in patients with COPD²⁰. Using a constrained linear mixed model adjusted for sex and baseline CAT score, it was estimated that the constrained baseline geometric mean of IL-8 was 1.02 ng ml^{-1} (95% confidence interval (CI): $0.65\text{--}1.61 \text{ ng ml}^{-1}$) and changed to 0.55 ng ml^{-1} (95% CI $0.32\text{--}0.96 \text{ ng ml}^{-1}$) and 1.16 ng ml^{-1} (95% CI $0.72\text{--}1.87 \text{ ng ml}^{-1}$) in the NR and placebo groups, respectively, after the 6-week treatment period. The least squares mean change from baseline in sputum IL-8 was -46.2% (95% CI -69.5% to -5.2%) in the NR group and 13.4% (95% CI -25.9% to 73.6%) in the placebo group, with an estimated treatment difference of -52.6% (95% CI -75.7% to -7.6% ; $P = 0.030$) (Fig. 2b). In post hoc analyses, this effect persisted until the follow-up 12 weeks after the end of treatment (estimated treatment difference, -63.7% ; 95% CI -85.7% to -7.8% ; $P = 0.034$). However, these findings should be interpreted with caution due to the small sample size and large CIs. As a result of the randomization of study participants, baseline levels of IL-8 in patients with COPD were higher in the NR group compared to the placebo group (Fig. 2b and

Table 1 | Baseline characteristics of the study populations

	COPD			Lung-healthy control		
	Total	Placebo	NR	Total	Placebo	NR
	(n=40)	(n=20)	(n=20)	(n=18)	(n=8)	(n=10)
Age, years	71.9±5.8	71.7±6.2	72.2±5.5	70.9±6.4	72.0±7.0	70.0±6.0
Female sex, n (%)	25 (62.5)	14 (70.0)	11 (55.0)	12 (66.7)	6 (75.0)	6 (60.0)
Weight, kg	76.4±16.8	79.7±12.9	73.2±19.8	72.0±11.3	70.5±12.1	73.2±11.1
Height, m	1.68±0.09	1.70±0.10	1.67±0.09	1.69±0.09	1.65±0.09	1.73±0.07
BMI, kg m ⁻²	26.9±4.8	27.8±4.4	26.0±5.2	25.0±2.8	25.7±2.7	24.4±2.8
FEV ₁ , % of predicted	64.5±18.4	63.1±15.9	65.9±20.9	117.4±15.5	113.8±15.4	120±15.7
FVC, % of predicted	95.5±24.0	93.4±23.6	97.6±24.9	116.3±15.5	112.9±14.9	119.0±16.2
FEV ₁ /FVC	52.5±11.6	53.3±13.1	51.7±10.3	78.3±4.0	78.3±3.5	78.4±4.6
Pack-years	34.4±13.4	32.3±12.0	36.6±14.8	N/A	N/A	N/A
Current medication, n (%)						
LABA+LAMA	22 (55.0)	11 (55.0)	11 (55.0)	N/A	N/A	N/A
LABA	3 (7.5)	2 (10.0)	1 (5.0)	N/A	N/A	N/A
LAMA	7 (17.5)	3 (15.0)	4 (20.0)	N/A	N/A	N/A
SABA	16 (40.0)	5 (25.0)	11 (55.0)	N/A	N/A	N/A
ICS	0 (0.0)	0 (0.0)	0 (0.0)	N/A	N/A	N/A

Data are presented as arithmetic means±s.d. or n (%). ICS, inhaled corticosteroid; LABA, long-acting β_2 -agonist; LAMA, long-acting muscarinic antagonist; N/A, not applicable; SABA, short-acting β_2 -agonist.

Supplementary Table 1), which may confound the response to the NR treatment. We did, however, find a negative relationship between baseline levels of IL-8 and pre-to-post changes in IL-8 after the intervention in the group receiving NR ($r = -0.77$, $P = 0.04$; Extended Data Fig. 1), which may suggest that NR has a stronger effect in individuals with higher baseline inflammation.

Secondary outcome NAD⁺ in whole blood

Our secondary outcome was to assess whether NR supplementation could augment NAD⁺ levels to a similar extent in both patients with COPD and lung-healthy controls. Blood samples for assessing NAD⁺ levels were collected in the fasted state, and participants took the last dose of NR/placebo 12–14 h before the post-intervention assessments to limit the effect of acute supplementation²¹. Six weeks of oral supplementation with NR increased NAD⁺ levels in whole blood by 71.1 μM (95% CI 57.2–85.0 μM) in patients with COPD and by 49.4 μM (95% CI 30.9–68.0 μM) in lung-healthy controls, with no change for placebo (Fig. 2c). NAD⁺ levels had completely returned to baseline levels at the follow-up assessment 12 weeks after the end of treatment (Fig. 2d and Supplementary Table 2). Baseline NAD⁺ levels were lower in patients with COPD compared to lung-healthy controls (least squares mean 31.9 μM (95% CI 30.3–33.5 μM) versus 34.8 μM (95% CI 32.4–37.1 μM), respectively; Fig. 2c) and was, in post hoc analyses, positively associated with lung function (Extended Data Fig. 2a), which suggests that NAD⁺ levels decrease with increasing disease severity. It has been suggested that a greater NAD⁺ response to NR supplementation occurs in individuals with lower baseline levels of NAD⁺ (ref. 11). Our findings only partly support this hypothesis because, although patients with COPD did experience a numerically greater increase in NAD⁺ levels after treatment compared to controls—with an estimated difference of 18.8 μM (95% CI –4.4 μM to 42.0 μM ; $P = 0.11$)—there was no relationship between baseline NAD⁺ levels and the response to 6 weeks of NR supplementation (Extended Data Fig. 2b).

Secondary outcome IL-6 in plasma

The anti-inflammatory effect of boosting NAD⁺ levels was shown in previous studies in humans^{22,23}. A further secondary outcome was,

therefore, to investigate the impact of NR on inflammation in both circulating and airway compartments. We measured plasma levels of the SASP cytokine IL-6 and found higher levels in patients with COPD compared to lung-healthy controls, but no effects of NR treatment could be detected (Fig. 2d). Because IL-8 is known to induce neutrophil chemotaxis, we conducted post hoc analyses and found an estimated treatment difference in sputum neutrophil differential count of 58% (95% CI –78% to –17%; $P = 0.009$; Extended Data Fig. 3). Sputum IL-6, neutrophil elastase and the number of macrophages remained unchanged compared to placebo (Extended Data Fig. 3).

Exploratory outcome epigenetic age in peripheral blood mononuclear cells

COPD has been described as a disease of accelerated lung aging², and NAD⁺ has been implicated in regulating epigenetic processes, such as DNA methylation²⁴. We, therefore, explored epigenetic age as measured by DNA methylation changes in peripheral blood mononuclear cells (PBMCs)^{25–28}. Additionally, we explored ‘System age’, which aims to capture epigenetic aging in distinct physiological systems²⁹. After treatment with NR, we observed a general pattern suggesting a decrease in epigenetic age in patients with COPD (Fig. 3a). Rate of aging was reduced after NR as calculated by the Horvath clock ($P = 0.024$), but this change did not differ from placebo ($P = 0.16$). In post hoc analyses at the 12-week follow-up, we observed a pattern of reduced rate of aging in the NR group for ‘Index’ ($P = 0.027$) but also for the ‘System age’ clocks Metabolic ($P < 0.0001$), Brain ($P = 0.0002$) and Liver ($P = 0.010$), albeit with no differences from placebo ($P = 0.28$, $P = 0.14$, $P = 0.15$ and $P = 0.36$, respectively; Extended Data Fig. 4). No clear pattern of NR treatment was seen for lung-healthy controls (Supplementary Table 4). Across numerous clocks, we observed indications of accelerated aging in patients with COPD when compared to lung-healthy controls (Fig. 3a).

Exploratory outcome RNA sequencing in nasal epithelial cells

To gain mechanistic understanding of how NR could impact the airway epithelium, we used RNA sequencing to explore changes in global gene expression of nasal epithelial cells. NAD⁺ regulates DNA repair³⁰, and NR supplementation led to an upregulation of gene sets related to

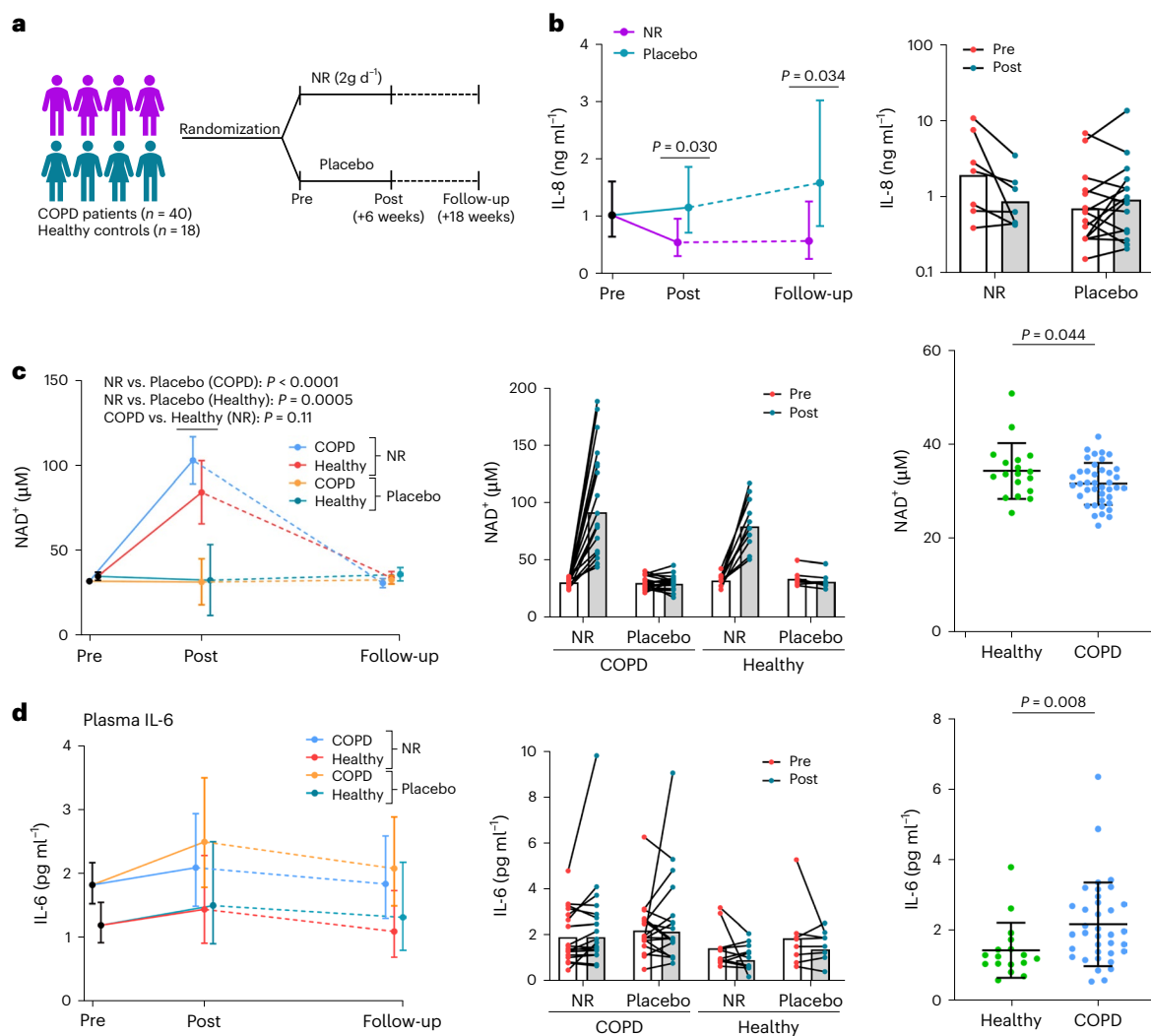


Fig. 2 | Augmented NAD⁺ levels after NR supplementation in COPD decrease airway inflammation. **a**, Study design. **b**, Changes in the primary outcome IL-8 in sputum from baseline (Pre, *n* = 30) after 6 weeks of NR/placebo supplementation (Post, *n* = 26) and at follow-up after 12 weeks (Follow-up, *n* = 24), along with individual values in patients with COPD: paired pre-intervention to post-intervention sputum samples where available are shown (*n* = 22). A constrained linear mixed model adjusted for sex and baseline CAT score was used for analysis on log-transformed data, and back-transformed least square means $\pm$ 95% CI with two-sided *P* values are shown. **c**, Changes in the secondary outcome whole-blood NAD⁺ levels from baseline (Pre, COPD: *n* = 40, Healthy: *n* = 18) to 6 weeks (Post, COPD: *n* = 37, Healthy: *n* = 18) and at follow-up after 12 weeks (Follow-up, COPD: *n* = 35, Healthy: *n* = 18), along with individual values in patients with COPD

and lung-healthy controls with paired pre-intervention to post-intervention whole-blood samples (COPD: *n* = 37, Healthy: *n* = 18) and NAD⁺ levels at baseline (mean $\pm$ s.d.). A constrained linear mixed model adjusted for sex was used for analysis, and least square means $\pm$ 95% CI with two-sided *P* values are shown. **d**, Changes in the secondary outcome plasma IL-6 levels from baseline (Pre, COPD: *n* = 40, Healthy: *n* = 18) to 6 weeks (Post, COPD: *n* = 36, Healthy: *n* = 18) and at follow-up after 12 weeks (Follow-up, COPD: *n* = 35, Healthy: *n* = 18), along with individual values in patients with COPD and lung-healthy controls with paired pre-intervention to post-intervention whole-blood samples (COPD: *n* = 36, Healthy: *n* = 18) and NAD⁺ levels at baseline (mean $\pm$ s.d.). A constrained linear mixed model adjusted for sex was used for analysis, and least square means $\pm$ 95% CI with two-sided *P* values are shown. Corrections for multiplicity were not made.

the maintenance of genomic integrity in both lung-healthy controls and patients with COPD (Fig. 3b). DNA damage leads to increased inflammation³¹, and COPD is characterized by an aberrant immune response to triggers, such as viruses³². Notably, NR supplementation led to a downregulation of gene sets related to immunity and antigen processing in both patients with COPD and lung-healthy controls. A full list of upregulated and downregulated gene sets can be found in Supplementary Table 8. Strikingly, 12 weeks after treatment, gene sets related to immunity were still suppressed in patients with COPD, whereas these changes had been normalized in controls (Extended Data Fig. 5). NAD⁺ is known to stimulate mitochondrial biogenesis³³, and NR treatment upregulated gene sets related to cellular respiration in the airways of patients with COPD at follow-up (Extended Data Fig. 5c). Principal component analysis indicated that the main effect of NR treatment occurred

on principal component 1 (PC1, *x* axis) in patients with COPD and on PC2 in lung-healthy controls (Extended Data Fig. 6a). Notably, female post-menopausal smokers are at increased risk of COPD³⁴, and, in the entire unselected gene set, we observed considerable sexual dimorphism (Extended Data Fig. 6b). Collectively, these findings suggest that NR may alter the COPD-related transcriptome toward increased DNA repair, reduced inflammation and increased cellular respiration.

Post hoc outcome senescence prediction in sputum

Cellular senescence is a complex process characterized by irreversible cell cycle arrest and is considered an important driver of COPD through the accumulation of senescent cells and activation of the SASP³⁵. To explore whether supplementation with NR would affect cellular senescence, we used a deep-learning-based approach³⁶ to predict cellular

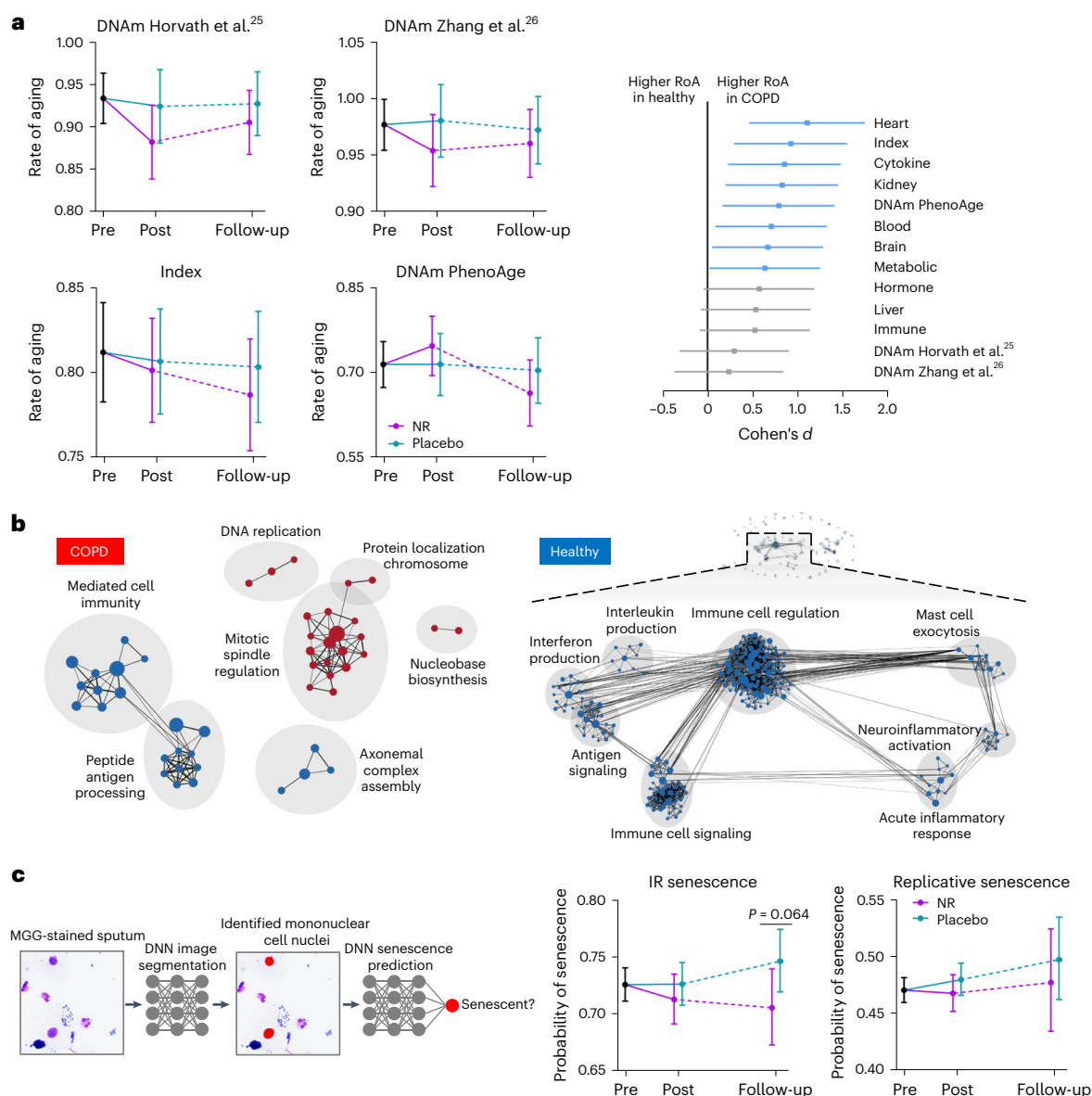


Fig. 3 | NR treatment effects on epigenetic aging, transcription signaling and predicted cellular senescence in patients with COPD. a, Change in the exploratory outcome rate of aging (RoA: epigenetic age/chronological age) from baseline (Pre, $n = 36$) after 6 weeks of NR/placebo supplementation (Post, $n = 31$) and after a 12-week follow-up (Follow-up, $n = 25$) in patients with COPD. Four different epigenetic clocks are shown. A constrained linear mixed model adjusted for sex and baseline CAT score was used for analysis showing least square means $\pm$ 95% CI. Forest plot showing differences in RoA between patients with COPD ($n = 36$) and lung-healthy controls ($n = 15$) measured in PBMCs. Each line represents the mean $\pm$ 95% CI effect size (Cohen's *d*) for different epigenetic clocks, with positive values indicating higher RoA in patients with COPD. Blue color indicates non-overlap of 95% CI with zero. **b,** Exploratory outcome RNA

sequencing in nasal epithelial cells. RNA was extracted and sequenced to generate GSEA maps showing functional groups (shaded circles) of gene sets detected ($FDR < 0.05$) in patients with COPD and lung-healthy controls after 6 weeks of NR treatment. Node size indicates gene set size, and color indicates direction of change, with red being upregulated and blue being downregulated with NR treatment. **c,** Analysis workflow showing a representative micrograph of a sputum sample and monocyte nuclei detection by the DNNs. Change in the post hoc outcome predicted IR-induced senescence and replicative senescence from baseline (Pre, $n = 29$) after 6 weeks of NR/placebo supplementation (Post, $n = 25$) and after a 12-week follow-up (Follow-up, $n = 24$) in patients with COPD. A similar statistical model as **a** was used. Corrections for multiplicity were not made.

senescence using the nuclear morphology of mononuclear cells in the sputum of patients with COPD (Fig. 3c). This approach can distinguish between cellular senescence caused by radiation and by replicative exhaustion. We found that 6 weeks of supplementation with NR did not affect levels of senescence based on nuclear morphology; however, 12 weeks after the supplementation period, we observed a pattern of decreased predicted cellular senescence in the NR group for ionizing radiation (IR)-like senescence compared to placebo (Fig. 3c and Supplementary Table 3). Notably, in post hoc correlation analyses, cellular senescence was positively related to the SASP factors IL-8 and IL-6

(Extended Data Fig. 7a), lending evidence toward the use of this method in sputum samples. Moreover, epigenetic age was positively related to both airway inflammation and cellular senescence, but changes in epigenetic age over the course of the trial were not correlated with either cellular senescence or airway inflammation (Extended Data Fig. 7b). Given that our findings *ex vivo* suggested that NR could impact cellular senescence, we tested the effects of NR in cells in culture. We found that NR maintained NAD^+ levels and increased survival after oxidative stress in immortalized airway epithelial cells (Extended Data Fig. 8a). Furthermore, NR decreased predicted senescence in a dose-dependent manner

after IR-induced damage in primary fibroblast cell lines, with a similar pattern seen for predicted senescence after UV damage (Extended Data Fig. 8b). Altogether, this suggests that NR may reduce senescence *ex vivo* and *in vitro*, consistent with recent findings^{37,38}. However, caution is warranted as key markers of cellular senescence were not comprehensively evaluated in this study.

We observed heightened levels of leukocytes in patients with COPD compared to lung-healthy controls but no effect of treatment with NR (Supplementary Table 5). Regarding physiological outcomes, there were no effects of NR supplementation on lung function (Supplementary Table 6) or on symptom severity (Supplementary Table 7); however, we did not expect to observe this given the short trial time and the patient population consisting of patients with COPD in stable conditions.

Discussion

In this randomized, placebo-controlled, double-blind clinical trial, we investigated the effects of NR supplementation on airway inflammation in patients with COPD. Although the size of this mechanistic study precluded assessment of the effect on clinical outcomes, our findings suggest that NR may be able to reduce lung inflammation and, thereby, reduce the risk of exacerbations and improve clinical outcomes. Other studies targeting neutrophilic inflammation and related pathways have suggested potential effects on symptoms, lung function and exacerbation risk³⁹. However, at this point, the exact immune mechanisms required to be targeted to achieve clinical effects remain unclear. Reducing the hyperinflammatory immune response in the airways of patients with COPD may require targeting of several immune mechanisms simultaneously to rebalance into immune homeostasis. Future research needs to explore the effect of NR treatment in response to airway pathogens as well as clinical effects in larger studies.

A key strength of this study is the randomized, placebo-controlled design and the inclusion of a lung-healthy control group. However, there are limitations that need to be addressed. It was not possible to collect paired sputum samples from all patients due to the well-known difficulty that some individuals have with expectoration⁴⁰. As a result, the primary outcome should be interpreted with caution, acknowledging the reduced sample size compared to our original power calculation and its potential impact on the study's statistical power. The coronavirus disease 2019 (COVID-19) pandemic constrained the study, resulting in the loss of some of the study assessments, including other markers of inflammation and NAD⁺ metabolomics. The short treatment period of 6 weeks and the small sample size could explain why no effects were seen on symptom severity in patients with COPD. Indeed, to detect clinically meaningful changes in the CAT score, the required sample size would be approximately 130 total patients given the observed variance. To measure cellular senescence, we applied a deep neural network (DNN) shown to detect senescent cells with up to 95% accuracy³⁶. We acknowledge that our results need to be confirmed and replicated in longer-term trials with larger sample sizes, applying multiple methods to assess cellular senescence. Despite the small sample size, the randomized design and the inclusion of patients with COPD with varying disease severity support the generalizability of the trial findings to the broader COPD population, with the minimally invasive intervention further enhancing its applicability. Lastly, given that numerous tests of secondary hypotheses are being performed, each at a 0.05 significance level, there is a considerable chance that at least one test will yield significant results. No corrections for multiplicity were made, and secondary analyses must be interpreted with caution.

Our study demonstrates that supplementation with NR reduced airway inflammation and increased NAD⁺ levels in whole blood. In exploratory analyses, treatment with NR showed indications of upregulated gene pathways related to genomic integrity in the airways and reduced epigenetic aging, possibly through a reduction in cellular senescence. These exploratory analyses need to be confirmed in future

trials. Furthermore, NR was demonstrated to be safe and well tolerated at 2 g d⁻¹ in both lung-healthy controls and patients with COPD. Our findings suggest that NR could possibly be a viable treatment option for patients with COPD.

Methods

Ethical approval, informed consent and study location

All procedures were approved by the regional ethical review board in Copenhagen (H-20074975). All participants provided written informed consent before enrollment into the study, consistent with the principles outlined in the Declaration of Helsinki. The clinical examinations were conducted between 5 August 2021 and 18 August 2022 at the Respiratory Research Unit at Bispebjerg Hospital, Copenhagen, Denmark. The study was registered on <https://www.clinicaltrials.gov/> under the identifier [NCT04990869](https://www.clinicaltrials.gov/ct2/show/study?term=NCT04990869). The trial followed CONSORT and SPIRIT guidelines. The Study Protocol and the Statistical Analysis Plan are included in Supplementary Information.

Study participants

Forty patients with a diagnosis of COPD (post-bronchodilator forced expiratory volume after 1 s (FEV₁)/forced vital capacity (FVC) < 0.7) were enrolled in this study (Table 1). Inclusion criteria included age of at least 60 years, BMI between 18.5 kg m⁻² and 40.0 kg m⁻², weight of at least 40 kg at enrollment, a smoking history of at least 10 pack-years but currently ex-smokers within 6 months, no use of inhaled corticosteroids, reported experiencing worsening of symptoms in relation to respiratory infections and a blood eosinophil count of less than 0.3 × 10⁹ cells per liter. Exclusion criteria included having had an exacerbation of COPD or severe airway infection within the last 2 months, chronic use of supplements containing NR or vitamin B before and throughout the trial or a cancer diagnosis within 5 years. Furthermore, we enrolled a convenience sample of lung-healthy controls who were similar to the patients with COPD in terms of age, sex and BMI but who reported being never-smokers with no history of lung disease (Table 1). Two lung-healthy controls were randomized to the placebo group but were later diagnosed with asthma and were, therefore, excluded from further analyses.

Study design, randomization and intervention

This was a single-center, double-blind, placebo-controlled clinical trial with a 6-week intervention phase and a 12-week follow-up period. The intervention consisted of ingesting 2 g of NR or placebo for 6 weeks (four 250-mg capsules consumed with meals in the morning and evening). Participants attended a screening visit and were randomized only if all inclusion criteria and none of the exclusion criteria were fulfilled. All participants were allocated 1:1 to receive NR or placebo using permuted block randomization with a block size of four. Independent blocks were generated for COPD and control, and patients with COPD were additionally randomized and stratified by CAT score (0–15 and 16–40). Randomization was performed by a member of the study team not involved in the assessment of outcomes. The study participants and members of the study team involved in the collection and analysis of the outcomes were blinded to the treatment condition. The placebo capsules had the same size, color, smell and appearance as the active drug tablets. The study visits included pre-treatment (week 1), post-treatment (week 6) and follow-up (week 18) assessments, for which participants arrived in the morning after an overnight fast. Participants took the last dose of NR/placebo 12–14 h before the post-intervention assessments. During each of the study visits, venous blood, sputum and nasal brush samples were collected. Lung function was measured using spirometry according to standards set by the American Thoracic Society/European Respiratory Society⁴¹. Venous blood samples were collected into Vacutainer tubes. Leukocyte count was analyzed using standardized clinical assays at Bispebjerg Hospital. Blood for NAD⁺ quantification was collected into sodium citrate tubes

and immediately placed on wet ice. Aliquots of 0.1 ml of blood were added to cryotubes containing 1 ml of 0.5 M perchloric acid, gently resuspended and stored at -80°C for later NAD^{+} quantification. AEs were recorded during the post-intervention and follow-up assessments.

Characteristics and questionnaires

Participant characteristics, including age, height, weight, comorbidities, prescribed medication, smoking history and alcohol consumption, were assessed during the pre-intervention assessments, in addition to a diagnostic chest X-ray in the patients with COPD only. Changes in weight and medications were noted in subsequent assessments. Symptom severity was measured in the patients with COPD using the CAT¹⁶ and the St. Georges Respiratory Questionnaire (SGRQ-C)⁴², and general health status was measured using the EuroQol 5-Dimension Questionnaire (EQ-5D-5L)⁴³.

Induced sputum

Sputum was induced in patients with COPD following standardized procedures. In brief, patients inhaled hypertonic saline (3%) using a nebulizer (EasyNeb 3, Fleam Nuova) for 5 min; if post-bronchodilator FEV_1 was less than 50% or less than 1.0 L, isotonic saline (0.9%) was used. Afterwards, patients rinsed their mouth with sparkling water, blew their nose and proceeded to expectorate sputum. This procedure was repeated three times, and lung function was monitored continuously. If FEV_1 fell by more than 20%, the test was stopped and a bronchodilator was administered. Sputum was expectorated in a 10-cm dish and kept at 4°C until processed within 2 h. Sputum plugs were collected in a 15-ml tube and weighed. The sample was diluted by adding eightfold of the sputum weight in milliliters of PBS and vortexed for 15 s. After 15 min on a bench rocker, the tube was centrifuged at 600g and 4°C for 10 min. A fourfold weight of the sputum weight in milliliters of dithiothreitol (DTT) (0.2%) was added, and the sample was vortexed for 15 s and put back on a bench rocker for 15 min. The sample was then filtered through a nylon filter washed in PBS, and cell viability was assessed with hemocytometry using trypan blue (0.4%). The sample was again centrifuged at 600g and 4°C for 10 min, and the supernatant was collected and stored at -80°C . The cell pellet was diluted in PBS for a cell concentration of 0.5×10^6 cells per milliliter, and cytopspins were prepared for differential cell counts in 400 non-squamous cells after staining cytopspin slides with May–Grünwald (Merck, 101424) and Giemsa (Merck, 109204) (MGG). Sputum slides used for senescence prediction were imaged in the Core Facility for Integrated Microscopy at Copenhagen University, using $\times 20$ magnification in an Axio Scan.Z1 automated slide scanner (Zeiss).

Nasal brushes

After participants cleaned their nose with isotonic saline, nasal brush samples were collected by inserting a brush (Gynobrush, Heinz Herenz Hamburg) approximately 4 cm into the left nostril toward concha media and rotating six times. The same procedure was repeated for the right nostril, and the brushes were put into a 15-ml Falcon tube containing 4 ml of PBS. The tube was vortexed for 20 s and centrifuged at 600g for 10 min at 4°C after discarding the brushes. The supernatant was discarded, and the cell pellet was lysed in 350 μl of RLT buffer (Qiagen) containing DTT (20 μl of 2 M DTT per 1 ml of RLT). RNA was extracted using a QIAcube (Qiagen) following the manufacturer's instructions, and samples were stored at -80°C .

Isolation of PBMCs

PBMCs were isolated from blood collected into EDTA-coated tubes. Blood was diluted (1:2) with PBS in 50-ml LeucoSEP (Greiner Bio-One) tubes containing Ficoll-Paque PLUS (Cytiva) separation medium and centrifuged at 1,000g for 15 min. Most of the plasma layer was removed, and the remaining sample was added to a new 50-ml Falcon tube, in which PBS to a total volume of 45 ml was added. The tube was

centrifuged at 300g for 10 min, and the supernatant was discarded. Any remaining red blood cells were lysed by adding ACK lysing buffer (Gibco, A1049201) for 3 min. The cells were washed three times in PBS, and the cell pellet was snap frozen in liquid nitrogen and stored at -80°C .

In vitro experiments

Cell culture. Human bronchial epithelial cells BEAS-2B (American Type Culture Collection (ATCC), CRL-9609) and primary human lung fibroblast cells MRC-5 (male, 14 weeks; ATCC, CCL-171) were maintained in Gibco DMEM medium, and primary human skin fibroblast cells (Coriell Institute), including AG08498 (male, 1 year), GM22159 (male, 1 d) and GM22222 (male, 1 d), were maintained in a 1:1 mixture of Gibco DMEM and F-12 media. All media were supplemented with 10% FBS (Sigma-Aldrich, F9665) and 100 U ml^{-1} penicillin–streptomycin (Gibco, 15140163), and cells were grown in 20% O_2 /5% CO_2 at 37°C .

Oxidative stress response. BEAS-2B cells were treated with increasing concentrations of menadione (Merck). Cells were seeded for 24 h in a six-well plate at a density of 10^6 cells per well. Cells were then treated with menadione for 4 h and subsequently recovered for 24 h. Cells were co-treated and left to recover with or without 1 mM NR (Elysium Health). Cells were washed two times with PBS, harvested by trypsin, resuspended in medium and spun at 500g and 4°C for 3 min. Cell pellets were resuspended, lysed in 0.2 M perchloric acid (HClO_4) and spun at 13,000g and 4°C for 2 min. Supernatant was diluted 200 $\times$ in 100 mM Na_2HPO_4 and stored at -80°C until later NAD^{+} quantification, whereas the cell pellets were used for protein quantification using a Pierce BCA Protein Assay Kit (Thermo Fisher Scientific). Cell survival was assessed by seeding cells in a 96-well plate at a density of 3,000 cells per well. After the same treatment and recovery as described above, cells were washed one time in PBS and incubated at room temperature in paraformaldehyde for 15 min. Cells were washed two times in PBS and stained with DAPI (AppliChem) for 10 min at room temperature in the dark. After 1 $\times$ wash in PBS, cells were stored in PBS in the dark at 4°C until later fluorescent imaging using $\times 20$ magnification on an IN Cell Analyzer 2200 high-content microscope (GE Healthcare Life Sciences). Image analysis was done using IN Cell Analyzer 1000 workstation software (GE Healthcare Life Sciences). Nuclei were segmented based on the DAPI signal using the TopHat segmentation method, and the total number of cells per well was counted.

Senescence induction. Cellular senescence was induced by exposing primary fibroblast cells (MRC-5, AG08498, GM22159 and GM22222) to IR and UV-C (254 nm) at increasing intensities. Cells were seeded for 24 h in a 96-well plate at a density of 3,000 cells per well. After aspirating media from the wells, cells were exposed to UV-C irradiation (custom-built machine). Afterwards, media containing increasing concentrations of NR were added, and cells were incubated for 48 h. For IR, media were aspirated and replaced by fresh media containing NR. Cells were exposed to IR using an YXLON Smart Maxishot, and cells were subsequently incubated for 48 h. Cells exposed to IR or UV-C were DAPI stained and imaged as described above.

Sample analyses

NAD^{+} quantification. For human whole-blood lysates, samples were thawed on ice and centrifuged at 13,000g for 2 min, and the supernatant was diluted 200 $\times$ in 100 mM Na_2HPO_4 . Already diluted supernatant from cell culture experiments was thawed on ice. As described previously⁴⁴, a reaction mixture containing 5 μl of 5 mg ml^{-1} Resazurin, 10 μl of flavin mononucleotide, 26 μl of 50 U ml^{-1} diaphorase, 100 μl of 1 M nicotinamide, 200 mM Na_2HPO_4 , 200 μl of absolute ethanol, 1,200 μl of 750 U ml^{-1} alcohol dehydrogenase and 3,500 μl of water was prepared, and a 100- μl reaction mixture was added to 1 ml of the diluted supernatant. The samples were measured by repeated fluorescence

spectrometry (one readout per minute for 30 min with excitation at 544 nm and emission at 580 nm) on a Hidex Sense reader to estimate total NAD⁺ levels.

IL-8, IL-6 and neutrophil elastase in sputum. The concentration of IL-8 and neutrophil elastase in sputum supernatant was measured using ELISAs (R&D Systems). Manufacturer-reported assay ranges are 31.2–2,000 pg ml⁻¹ and 46.9–3,000 pg ml⁻¹ for IL-8 and neutrophil elastase, respectively; thus, samples were diluted 1:5 for IL-8 and 1:10 for neutrophil elastase in reagent diluent. The concentration of IL-6 in sputum supernatant was measured using an S-PLEX assay (Meso Scale Discovery), which has a reported assay range of 1.1–6,000 fg ml⁻¹. Samples were diluted 1:250. Samples were run in duplicate with a coefficient of variation of 5.0%, 4.1% and 2.1% for IL-8, IL-6 and neutrophil elastase, respectively.

IL-6 in serum. The concentration of IL-6 in serum was measured using the S-PLEX assay (Meso Scale Discovery), and samples were diluted 1:100.

Senescence prediction. Cellular senescence was predicted from nuclei shapes as previously described³⁶. We applied an image-based nuclear senescence predictor (NUSP), which was previously trained on human fibroblasts exposed to radiation or undergoing replicative exhaustion. This approach has been shown to have high accuracy in predicting senescence, correlation to numerous senescence markers and application to diverse cell types and species. Mononucleated nuclei were identified using U-NET segmentation, and the NUSP predicted senescence for identified nuclei based on morphology.

RNA sequencing in nasal epithelial cells. RNA sequencing was performed on 151 nasal epithelial cell samples collected by nasal brush as indicated above by BGI Genomics using whole RNA sequencing analysis on samples with an RNA integrity number (RIN) mean of 8.0 (range, 5.2–9.5). Paired-end reads were aligned to mm9 using Bowtie 2 (ref. 45). Differential expression analysis was performed using Salmon, tximeta and DESeq2 (ref. 46). Gene set enrichment analysis (GSEA) was performed on DESeq2-normalized counts, and comparisons were made against baseline values and between COPD and control. The gene set used was based on Gene Ontology terms. Terms were filtered for false discovery rate (FDR) < 0.05 and sorted by normalized enrichment score. Gene sets with FDR > 0.05 that were differentially expressed after treatment with NR (pre to post) and did not overlap with placebo were visualized using EnrichmentMap in Cytoscape (version 3.9.1). Similar gene sets were clustered together into functional groups using AutoAnnotate.

DNA methylation. PBMCs were shipped on dry ice and analyzed by Tempus Labs. DNA was isolated on a Maxwell RSC 48 instrument using the Maxwell RSC Blood DNA Kit (Promega, AS1400) according to the manufacturer's protocol. In brief, this kit uses cellulose-based binding of nucleic acids to provide a mobile solid phase that optimizes sample capture, washing and purification of gDNA. To ensure sufficient DNA for analysis, DNA was quantified on the Biomek i5 automated workstation using a SpectraMax Gemini spectrophotometer with a Quant-iT PicoGreen dsDNA kit (Thermo Fisher Scientific, P7589). Samples were then normalized on the Biomek i5 automated workstation. To run the sample on an Illumina microarray, 500 ng of DNA was bisulfite converted overnight using an EZ DNA Methylation Kit (Zymo Research). The bisulfite-treated DNA was amplified manually. A Freedom EVO workstation (Tecan Life Sciences) was used to automate fragmentation, purification, hybridization, washing and staining. DNA methylation profiles were measured using a custom Illumina 250K array developed by Elysium Health and Illumina. The custom array contains all Infinium I probe targets with additional probes that were chosen based on their technical robustness and to ensure representative coverage across

the genome. The custom 250K array was processed according to the manufacturer's instructions. Raw intensity data files were imported into the R environment (R version 4.1.2, <http://www.r-project.org>) using the 'minfi' package for pre-processing and analysis. Quality control of the methylation data was rigorously executed using the 'MethylAid', 'minfi' and 'ewastools' packages, ensuring stringent adherence to performance standards. 'MethylAid' assessed the median intensity of methylated versus unmethylated probes with a cutoff of 10, overall assay quality with a threshold of 12, bisulfite conversion with a limit of 11.75 and hybridization quality at 12.75, alongside detection *P* value requirements set to *P* = 0.05. 'ewastools' was used to evaluate 17 control metrics derived from specific control probes for each assay, as delineated in the BeadArray Controls Reporter Software Guide by Illumina, providing a comprehensive evaluation of assay performance. After quality control, background correction and normalization were performed using the minfi::preprocessnoob method, which is designed to adjust for optical noise and dye bias. Subsequently, the normalized data were converted into beta values representing the ratio of methylated probe intensity to the overall intensity. In estimating epigenetic age, we used epigenetic biomarkers derived from Elysium Health's proprietary algorithms, 'Index' and 'System age' clocks. The 'Index' clock leverages methodologies established in Higgins-Chen et al.²⁸, representing a refined second-generation epigenetic clock. The 'System age' clocks extend this approach across multiple physiological systems as outlined in Sehgal et al.²⁹. These were evaluated in conjunction with other epigenetic clocks, specifically those developed by Horvath et al.²⁵, Zhang et al.²⁶ and Levine et al.²⁷ (DNAm PhenoAge), to provide a comprehensive assessment of epigenetic aging.

Statistics

The primary outcome of this study was to determine whether changes from baseline in sputum IL-8 after NR supplementation were different from the corresponding changes after placebo treatment at 6 weeks. Previous work in older humans showed that 1 g of NR can reduce circulating inflammatory markers between 30% and 50% compared to placebo²². To determine whether NR could comparatively reduce airway inflammation, we estimated, based on previous data²⁰, that 10 participants per group would be needed to detect a between-group difference of 8 ng ml⁻¹ in IL-8, equating to a 40% reduction from a baseline of 21 ng ml⁻¹ and an s.d. of 5.74 ng ml⁻¹. These calculations were done using G*Power 3.1.9.2 and assume a 5% and 20% type I and type II error rate, respectively. Allowing for a 20% dropout and assuming conservatively that only 60% of patients with COPD would be able to expectorate sputum, we enrolled 20 participants into the NR group and 20 participants into the placebo group. To increase statistical power in explorative outcomes, patients not able to expectorate sputum were not excluded from the trial. A convenience sample of 20 lung-healthy controls was recruited, and comparisons between COPD and healthy should be considered exploratory. After data collection, no datapoints were excluded.

The effect of NR on the change from baseline to week 6 and week 18 in the primary outcome sputum IL-8 was analyzed using a constrained linear mixed model with an unstructured covariance after log transformation. The model included visit and a visit-by-treatment interaction term to allow for the treatment effect to change at each visit while assuming that the baseline means are constrained to be the same across the treatment groups due to randomization⁴⁷. This analysis was conducted on an intention-to-treat basis, with no interim analysis performed and no predefined stopping guidelines. The same model was applied to explorative outcomes with log transformation as appropriate, ensuring that assumptions of normality (Shapiro–Wilk test) and equal variance (Levene's test) were met (see figure legends for details). Analyses were adjusted for sex additionally for baseline CAT score (used in the randomization scheme). The least square means for treatment groups are presented with the associated two-sided 95% CI and *P* value.

No corrections for multiplicity were made, and secondary analyses should be considered exploratory. All analyses were performed with SAS version 8.3 software (SAS Institute). Figures were made using GraphPad Prism version 10 (GraphPad Software).

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

Requests for data supporting the findings reported in this paper should be made to the corresponding author and will be reviewed individually. Individual participant-level raw data containing confidential or identifiable patient information are subject to patient privacy and cannot be shared. For RNA sequencing data, DESeq2-normalized sequence counts for each individual are available at <https://doi.org/10.5061/dryad.d2547d8b5> (ref. 48). Unique biological materials cannot be shared as they are restricted by participant consent.

Code availability

No original code was created for this study.

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Author contributions

Conceptualization: K.L.N., C.P. and M.S.-K. Methodology: K.L.N., M.B.E., I.H., L.M.A., L.L.E., N.D.-P., M.V.D., M.B., L.P., D.S., R.W.D., A.S., J.T.T., S.B.D., C.P. and M.S.-K. Software: M.B.E., I.H. and D.S. Investigation: K.L.N., M.B.E., I.H., L.M.A., L.L.E., N.D.-P., M.V.D., J.T.T., S.B.D., C.P. and M.S.-K. Funding acquisition: K.L.N., A.S., J.T.T., C.P. and M.S.-K. Supervision: J.T.T., S.B.D., C.P. and M.S.-K. Writing—original draft: K.L.N., C.P. and M.S.-K. Writing—review and editing: K.L.N., M.B.E., I.H., L.M.A., L.L.E., N.D.-P., M.V.D., M.B., L.P., D.S., R.W.D., A.S., J.T.T., S.B.D., C.P. and M.S.-K.

Competing interests

D.S. and R.W.D. are employees of Elysium Health and own shares in the company. The other authors declare no competing interests.

Additional information

Extended data is available for this paper at <https://doi.org/10.1038/s43587-024-00758-1>.

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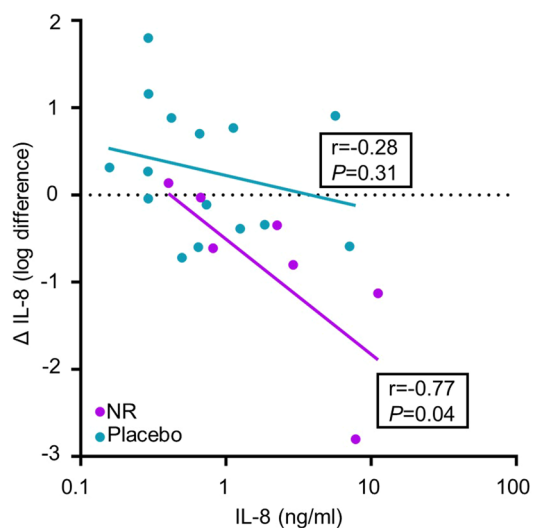
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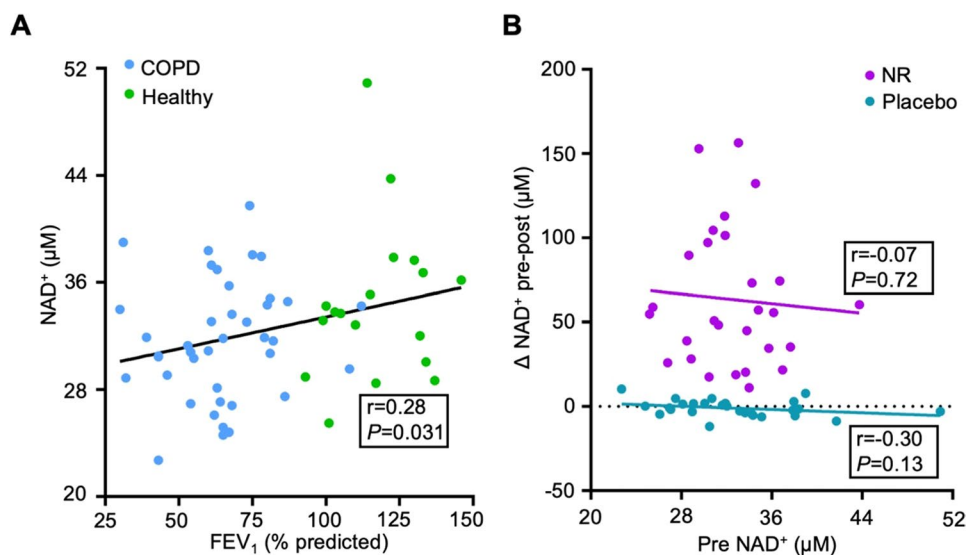
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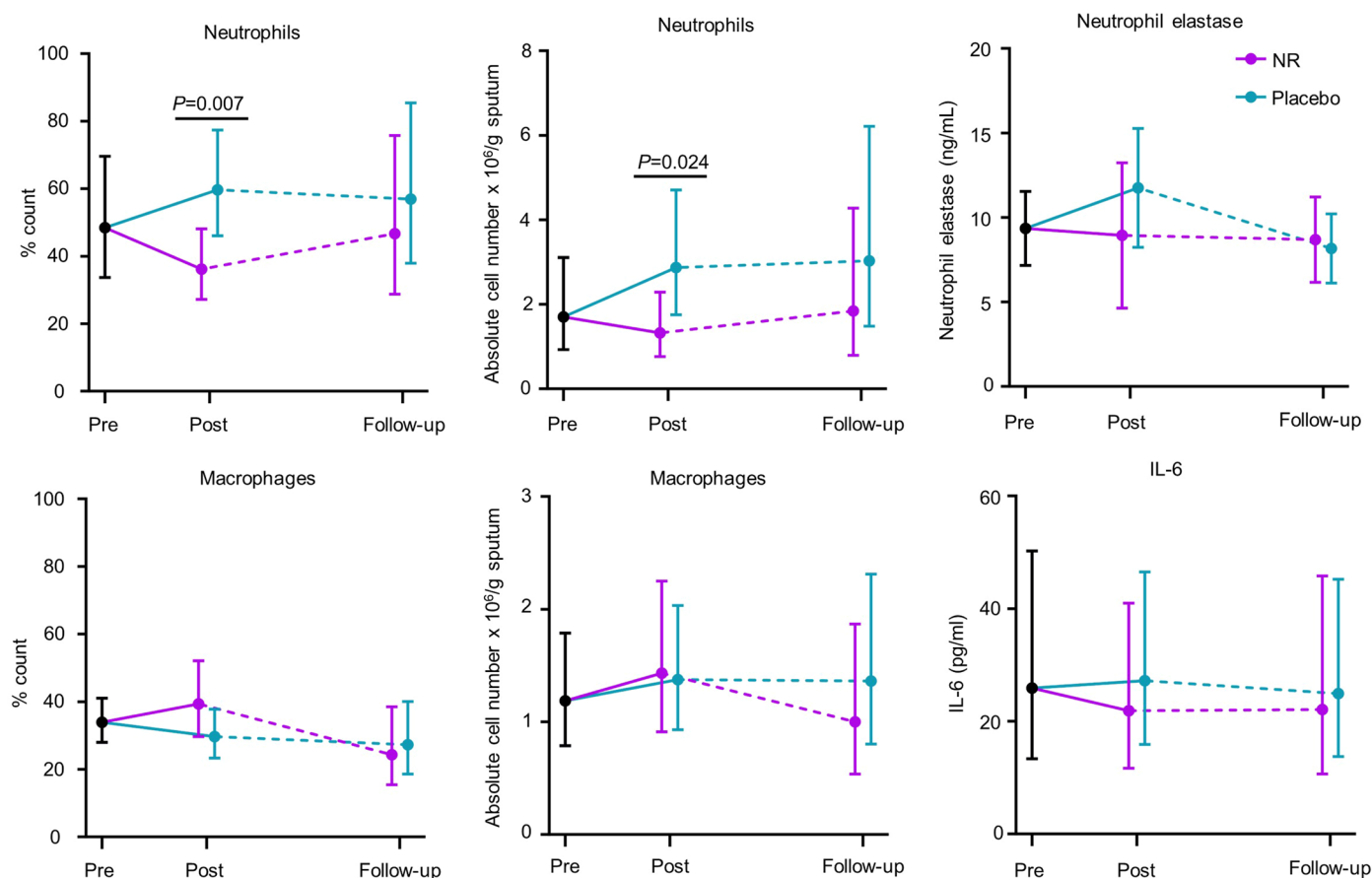
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Extended Data Fig. 1 | Correlation between baseline sputum IL-8 and the change in IL-8 following the intervention. Delta values are Pre- to post-intervention in the NR (n = 7) and placebo (n = 15) group of patients with COPD. Correlations are analyzed using Person's correlation on log-transformed data with two-sided P-values shown.

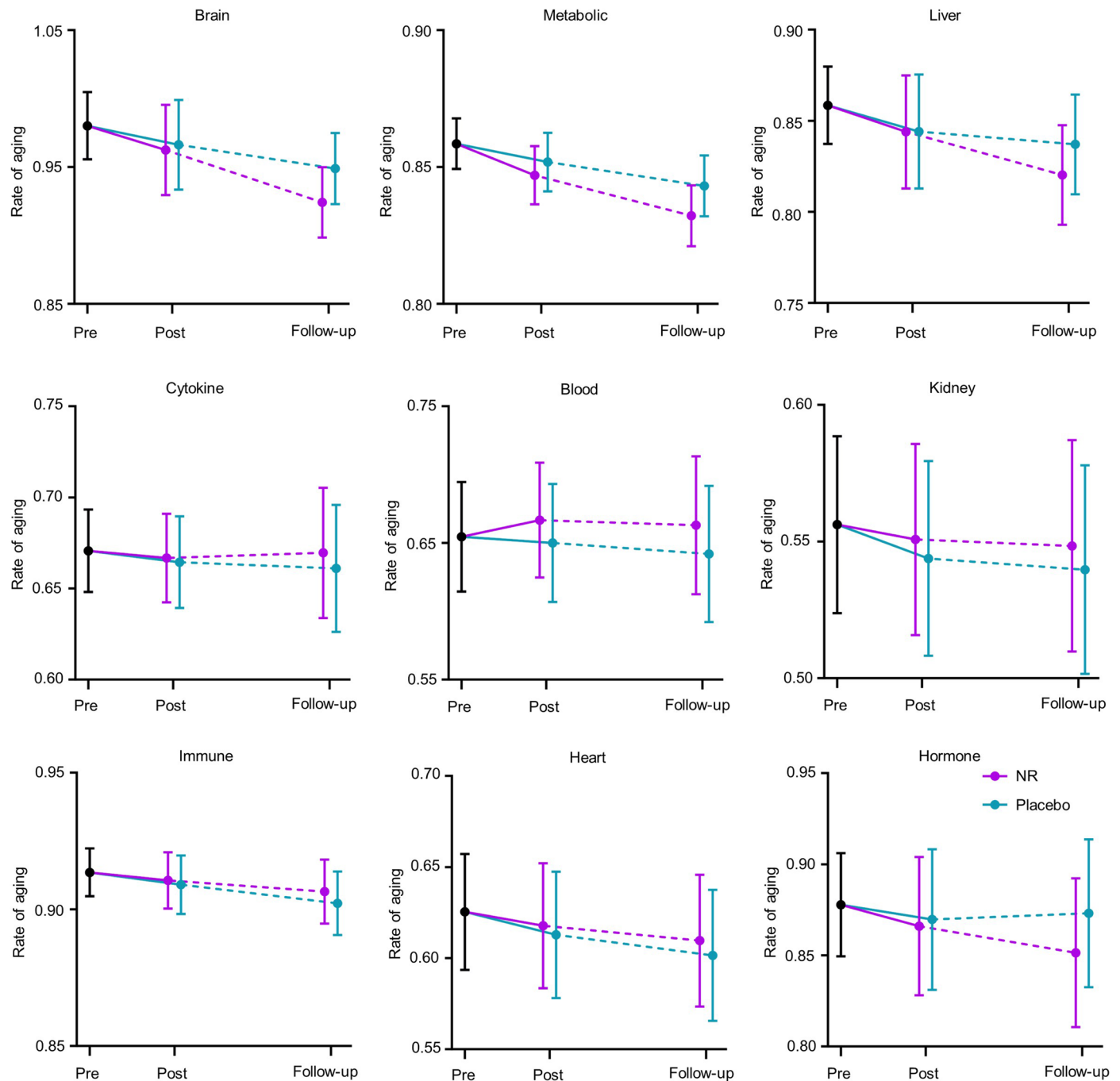


Extended Data Fig. 2 | Baseline correlations with NAD^+ . (a) Correlation between baseline NAD^+ and Forced Expiratory Volume (FEV_1) analyzed using Person's correlation. (b) Correlation between baseline NAD^+ and the change in NAD^+ following the intervention in pooled NR and placebo groups analyzed using Person's correlation.



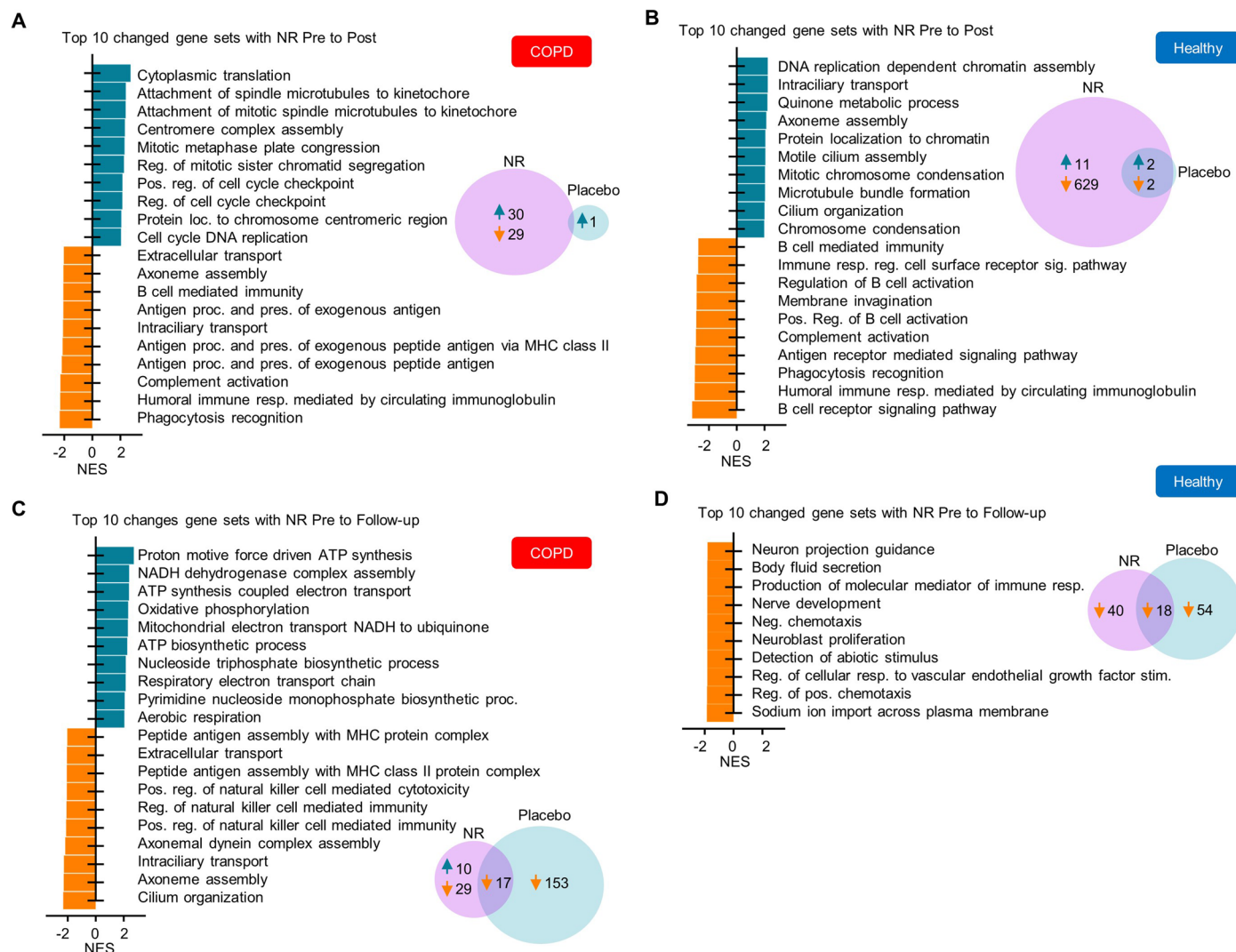
Extended Data Fig. 3 | Effects of NR treatment on sputum inflammatory markers. Changes in the post hoc outcomes sputum neutrophil and macrophage differential and absolute count, and neutrophil elastase and interleukin-6 (IL-6) levels from baseline (Pre, n = 30) following six weeks of NR/placebo supplementation (Post, n = 26) and at follow-up after

twelve-weeks (Follow-up, n = 24) in patients with COPD. A constrained linear mixed model adjusted for sex and baseline COPD assessment test (CAT) score was used for analysis on log-transformed data and back-transformed least square means $\pm$ 95% confidence intervals with two-sided *P*-values shown. Corrections for multiplicity were not made.



Extended Data Fig. 4 | Effects of NR treatment on epigenetic age. Change in rate of aging (epigenetic age/chronological age) from baseline (Pre, $n = 36$) following six weeks of NR/placebo supplementation (Post, $n = 31$) and after a twelve-week follow-up (Follow-up, $n = 25$) in patients with COPD. The nine different epigenetic

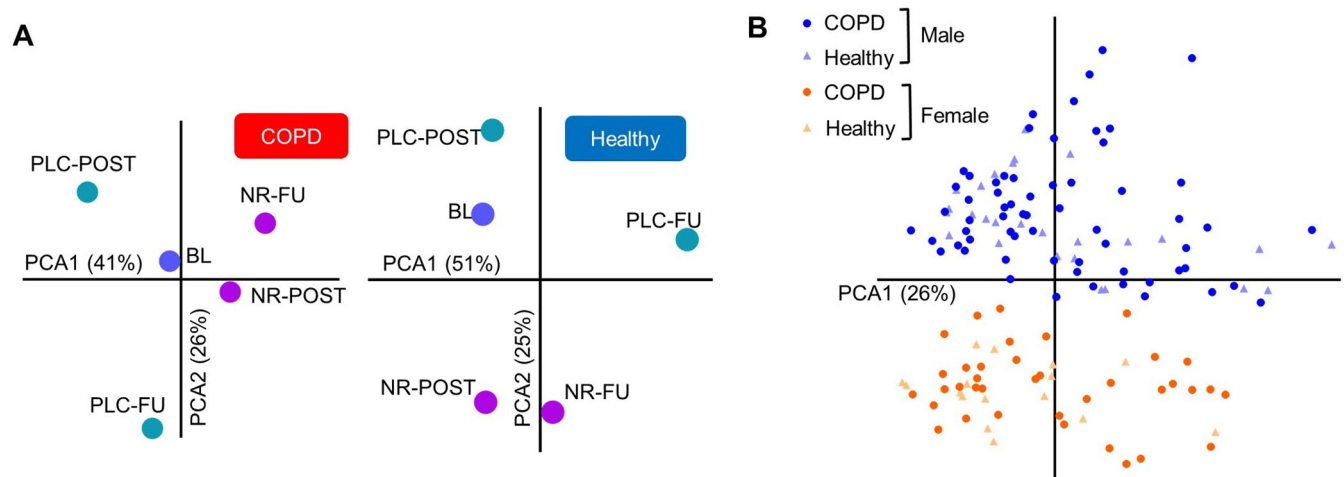
clocks included in the 'System age' clocks are shown. A constrained linear mixed model adjusted for sex and baseline COPD assessment test (CAT) score was used for analysis with least square means $\pm$ 95% confidence intervals with two-sided P -values shown. Corrections for multiplicity were not made.



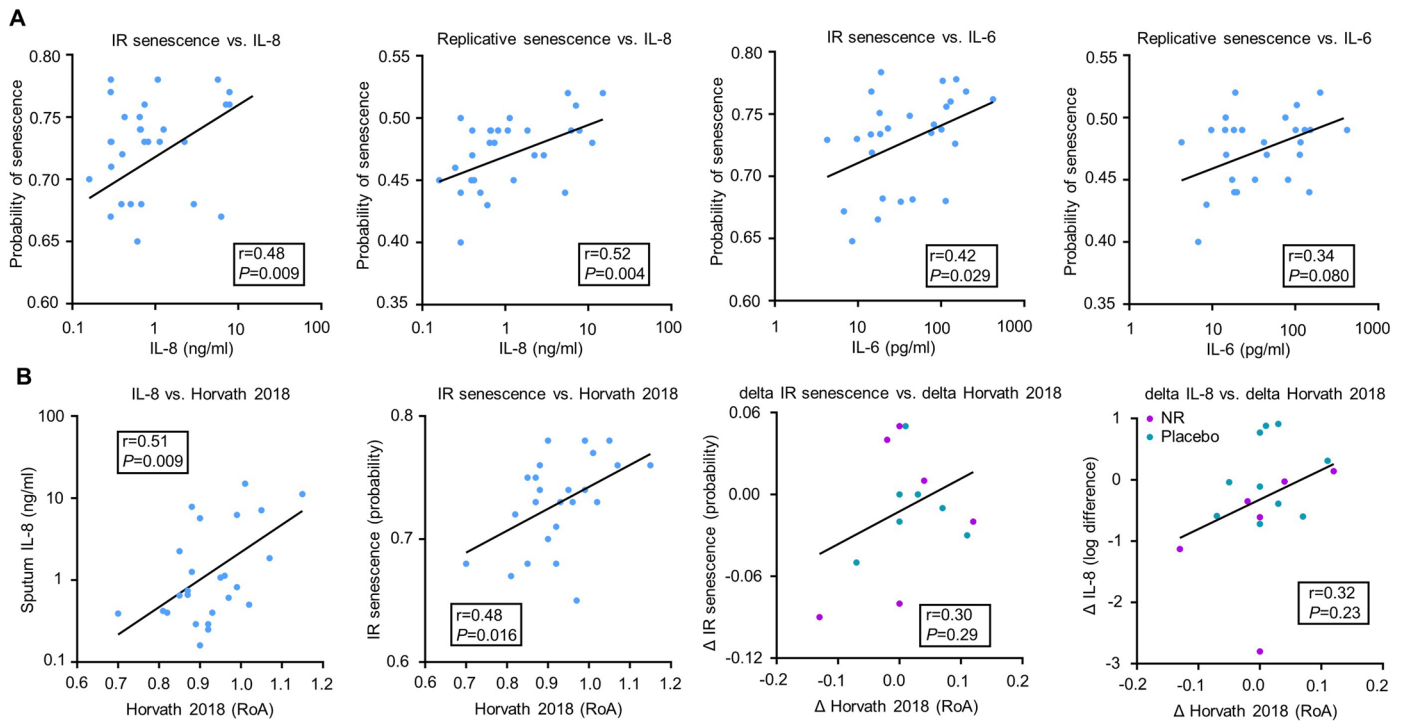
Extended Data Fig. 5 | Effects of NR treatment on gene set enrichment.

(a) Gene Set Enrichment Analysis (GSEA) of top 10 up- and down-regulated gene sets arranged by normalized enrichment score (NES) and Venn diagram of enriched gene sets following 6 weeks of NR supplementation in patients

with COPD and (b) lung-healthy controls. (c) Bar graph and Venn diagram of enriched gene sets after a 12-week follow-up period in patients with COPD and (d) lung-healthy controls.

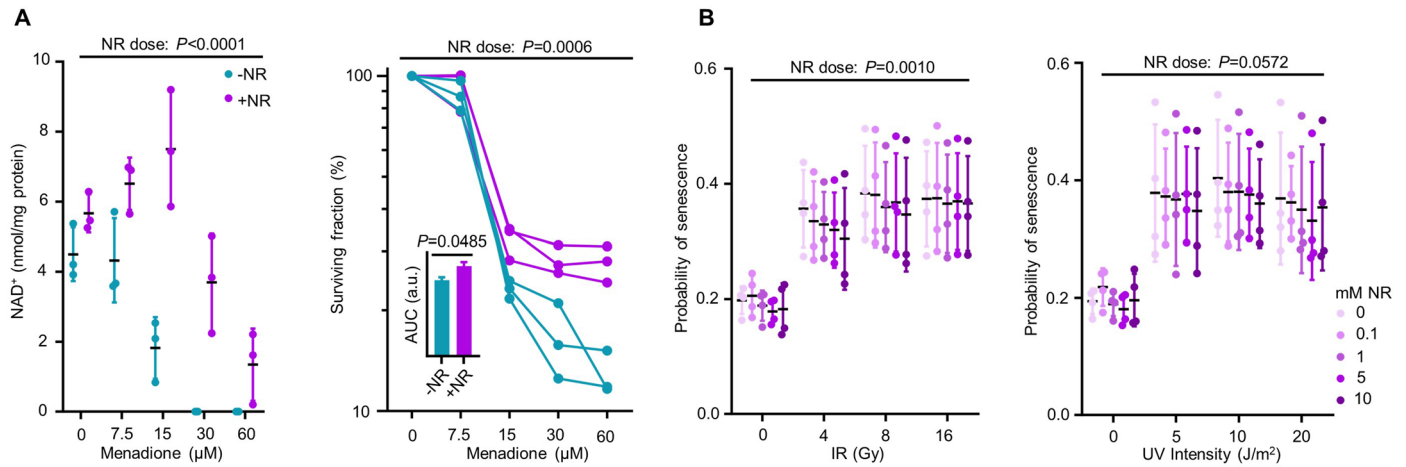


Extended Data Fig. 6 | Effects of NR treatment on principal component analysis. (a) Principal component analysis (PCA) of mean gene expression of nasal epithelial cells in patients with COPD and lung-healthy controls grouped into baseline (BL), post, and follow-up (FU) for NR and placebo (PLC) treated groups. **(b)** PCA of gene expression of individual subjects at a time point.



Extended Data Fig. 7 | Correlations between markers of inflammation, senescence and epigenetic age. (a) Correlation between baseline sputum IL-8 and predicted IR senescence ($n = 29$), sputum IL-8 and replicative senescence ($n = 29$), sputum IL-6 and predicted IR senescence ($n = 27$) and sputum IL-6 and replicative senescence ($n = 27$). (b) Correlation between baseline RoA and

sputum IL-8 ($n = 25$) and ionizing radiation-like senescence ($n = 25$). Correlation between change in RoA and ionizing radiation-like senescence ($n = 14$) and sputum IL-8 following the intervention ($n = 16$). Correlations were analyzed using Person's correlation on log-transformed data with two-sided P-values shown.



Extended Data Fig. 8 | Effects of NR *in vitro*. (a) NAD⁺ levels and cell viability in human bronchial epithelial cells (BEAS-2b) following exposure to increasing concentrations of menadione for 4 hours co-treated with or without 1 mM NR during exposure and 24 hours after exposure (mean $\pm$ standard deviation, $n = 3$ biological replicates). A main-effects analysis of variance (ANOVA) model was fitted to the data, and the two-tailed P -values for the effect of NR dose are shown. An unpaired t-test was used to test differences in area under the curve (AUC)

survival with or without NR with a two-tailed P -value shown. (b) Probability of senescence following exposure to ionizing radiation (IR) and ultraviolet light (UV-C) in primary human fibroblasts (MRC-5, AG08498, GM22159, GM22222) treated with increasing concentrations of NR for 24 hours after exposure (mean $\pm$ standard deviation, $n = 4$, with each cell line used as a biological replicate). A main-effects ANOVA model was fitted to the data, and the two-tailed P -values for the effect of NR dose are shown.

Extended Data Table 1 | AEs

	COPD				Lung-healthy			
	NR (n = 20)		Placebo (n = 20)		NR (n = 10)		Placebo (n = 8)	
	AE	Participants experiencing AEs	AE	Participants experiencing AEs	AE	Participants experiencing AEs	AE	Participants experiencing AEs
	N	N (%)	N	N (%)	N	N (%)	N	N (%)
Respiratory symptoms	5	4 (20.0)	3	3 (15.0)	0	0 (0.0)	0	0 (0.0)
Gastrointestinal symptoms	2	2 (10.0)	4	3 (15.0)	1	1 (10.0)	1	1 (12.5)
Pain	2	1 (5.0)	0	0 (0.0)	1	1 (10.0)	1	1 (12.5)
Fatigue	0	0 (0.0)	0	0 (0.0)	1	1 (10.0)	0	0 (0.0)
Renal and urinary symptoms	0	0 (0.0)	1	1 (5.0)	0	0 (0.0)	0	0 (0.0)
Reduced appetite	1	1 (5.0)	0	0 (0.0)	0	0 (0.0)	0	0 (0.0)
Increased hunger	0	0 (0.0)	0	0 (0.0)	2	2 (20.0)	0	0 (0.0)
Sleep issues	1	1 (5.0)	0	0 (0.0)	0	0 (0.0)	0	0 (0.0)
Itch	0	0 (0.0)	0	0 (0.0)	1	1 (10.0)	0	0 (0.0)
Dry mouth	1	1 (5.0)	0	0 (0.0)	0	0 (0.0)	0	0 (0.0)
Transient hypertension	0	0 (0.0)	1	1 (5.0)	0	0 (0.0)	0	0 (0.0)
Overall AEs	12	10 (50.0)	9	8 (40.0)	6	6 (60.0)	2	2 (25.0)

Two-tailed chi-square statistic of participants experiencing AEs in pooled NR group versus placebo group: 1.818, $P=0.178$. Data are presented as n (%).

Reporting Summary

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For all statistical analyses, confirm that the following items are present in the figure legend, table legend, main text, or Methods section.

n/a	Confirmed
☐	☒ The exact sample size (<i>n</i>) for each experimental group/condition, given as a discrete number and unit of measurement
☒	☐ A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly
☐	☒ The statistical test(s) used AND whether they are one- or two-sided <i>Only common tests should be described solely by name; describe more complex techniques in the Methods section.</i>
☐	☒ A description of all covariates tested
☐	☒ A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons
☐	☒ A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals)
☐	☒ For null hypothesis testing, the test statistic (e.g. <i>F</i> , <i>t</i> , <i>r</i>) with confidence intervals, effect sizes, degrees of freedom and <i>P</i> value noted <i>Give P values as exact values whenever suitable.</i>
☒	☐ For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings
☒	☐ For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes
☐	☒ Estimates of effect sizes (e.g. Cohen's <i>d</i> , Pearson's <i>r</i>), indicating how they were calculated

Our web collection on [statistics for biologists](#) contains articles on many of the points above.

Software and code

Policy information about [availability of computer code](#)

Data collection	All clinical data were captured using the electronic data capture system REDCap.
Data analysis	Statistical software was SAS software, version 8.3 (SAS Institute, Cary, NC, USA). Figures were made using Prism version 10 (GraphPad). Image analysis was done using INCell Analyzer 1000 workstation software (GE Healthcare Life Sciences). Quality control of the methylation data was rigorously executed using the 'MethylAid', 'minfi', and 'ewastools' packages.

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form of aggregate data summaries and via a data transfer agreement. Individual participant level raw data containing confidential or identifiable patient information is subject to patient privacy and cannot be shared. Unique biological materials cannot be shared as they are restricted by participant consent.

Research involving human participants, their data, or biological material

Policy information about studies with [human participants or human data](#). See also policy information about [sex, gender \(identity/presentation\), and sexual orientation](#) and [race, ethnicity and racism](#).

Reporting on sex and gender	Both male and female participants were included in this study. Demographic information about the number of males and females is provided in Extended Data Table 1.
Reporting on race, ethnicity, or other socially relevant groupings	The study uses ethnicity to estimate percentage expected lung function i.e. Global Lung Function Initiative reference equations. All participants in the present study were self-reported caucasian.
Population characteristics	Population characteristics is provided in Extended Data Table 1.
Recruitment	Subjects with a diagnosis of COPD were recruited from the Respiratory Medicine and Emergency departments at Bispebjerg Hospital, Denmark. Patients had agreed to being contacted regarding clinical trials before initial contact was made. Lung-healthy controls were recruited through advertisements on online websites, including Facebook. Since participation in the study was voluntary, self-selection bias could have influenced the recruitment process, potentially favoring the inclusion of healthier individuals. It is possible that larger effect sizes would be observed in a population with a higher proportion of sicker individuals.
Ethics oversight	All procedures were approved by the regional ethical review board in Copenhagen (H-20074975).

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☒ Life sciences ☐ Behavioural & social sciences ☐ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see nature.com/documents/nr-reporting-summary-flat.pdf

Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size	The primary outcome of the study was to determine whether changes from baseline in sputum IL-8 after NR supplementation were different from the corresponding changes after placebo treatment at 6 weeks. Previous work in older humans has shown that 1 g of NR can reduce circulating inflammatory markers between 30-50% compared with placebo. To determine whether NR could comparatively reduce airway inflammation, based on previous data we estimated that 10 participants per group would be needed to detect a between-group difference of 8 ng/ml in IL-8 equating to a 40% reduction from a baseline of 21 ng/ml and a SD of 5.74 ng/ml. These calculations were done using G*Power 3.1.9.2 and assume a 5% and 20% type I and type II error rate, respectively. Allowing for a 20% dropout and assuming conservatively that only 60% of patients with COPD would be able to expectorate sputum, we enrolled 20 participants into the NR group and 20 participants into the placebo group. Patients not able to expectorate sputum were not excluded from the trial to increase statistical power in secondary and explorative outcomes.
Data exclusions	Two lung-healthy control participants (n=2) were randomized to the placebo group but were later diagnosed with asthma and were therefore excluded from further analyses.
Replication	We did not replicate the experiments due to limited amounts of patient samples.
Randomization	All participants were allocated to receive NR or placebo using permuted block randomization with a block size of four. Independent blocks were generated for COPD and controls, and patients with COPD were additionally randomized stratified by CAT score (0-15 and 16-40). Randomization was performed by a member of the study team not involved in the assessment of outcomes.
Blinding	The study participants and members of the study team involved in the collection and analysis of the outcomes were blinded to the treatment condition. The placebo capsules had the same size, color, smell and appearance as the active drug tablets.

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☒	☐ Antibodies
☐	☒ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☒	☐ Animals and other organisms
☐	☒ Clinical data
☒	☐ Dual use research of concern
☒	☐ Plants

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging

Eukaryotic cell lines

Policy information about [cell lines and Sex and Gender in Research](#)

Cell line source(s)	The study includes human bronchial epithelial cells BEAS-2B (CRL-9609; ATCC, USA), primary human lung fibroblast cells MRC-5 (male, 14 weeks, CCL-171; ATCC, USA), primary human skin fibroblast cells (Coriell Institute, USA) including AG08498 (male, 1 year), GM22159 (male, 1 day), and GM22222 (male, 1 day).
Authentication	Cell lines were not authenticated.
Mycoplasma contamination	All cell lines were tested free of mycoplasma.
Commonly misidentified lines (See ICLAC register)	No commonly misclassified cell lines were used.

Clinical data

Policy information about [clinical studies](#)

All manuscripts should comply with the ICMJE [guidelines for publication of clinical research](#) and a completed [CONSORT checklist](#) must be included with all submissions.

Clinical trial registration	The study was registered at ClinicalTrials.gov (NCT04990869).
Study protocol	Refer to the supplemental material.
Data collection	The clinical examinations were conducted between 5 August 2021 and 18 August 2022 at the Respiratory Research Unit at Bispebjerg Hospital, Copenhagen, Denmark.
Outcomes	Pre-defined Primary outcome: Changes in sputum interleukin-8 levels measured using ELISA after 6 weeks of treatment with NR/placebo in patients with chronic obstructive pulmonary disease. Pre-defined Secondary outcomes: Changes in whole-blood NAD ⁺ levels measured using cyclic assay and in serum IL-6 levels measured using MSD after 6 weeks of treatment with NR/placebo in patients with chronic obstructive pulmonary disease and lung-healthy controls. Exploratory outcomes: Nasal epithelial cell transcriptomics (untargeted RNA sequencing) and DNA methylation in PBMCs (Elysium Index Chip) after 6 weeks in patients with chronic obstructive pulmonary disease and lung-healthy controls.

Plants

Seed stocks	N/A
Novel plant genotypes	N/A
Authentication	N/A